SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	UTeMku Cemerlang
Project / Solution Name	PaySah
Name (Leader)	Chong Kai Li
Email & Phone (Leader)	kailichong148@gmail.com | 0146312883
University Affiliation(s)	Universiti Teknikal Malaysia Melaka
Live Prototype URL	https://paysah.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/UTeMkuCemerlang/PaySah_Final

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

React/TypeScript presents order review and activity history. Dashboard, Telegram and Google Forms/Apps Script send order-linked receipts to Express. Private Replit App Storage holds images; PostgreSQL stores tenant-scoped orders, fields, jobs, policies, synthetic credits and audit records.

A persistent worker calls RunPod Serverless/vLLM with Qwen2.5-VL-7B-Instruct and Tesseract.js CPU reading/fallback. gpt-5.4-mini via KrackedDevs (litellm.rapidscreen.io) is used as a live advisory

third reader only when GPU and CPU disagree, and in an offline 36-image diagnostic. It cannot change fields, rules or payment; disabling it leaves the workflow operational.

Rules compare receipts with orders and synthetic bank records. PaySah warns and blocks by default. The Owner can override at their own responsibility with a reason and confirmation prompt; overrides are logged as MANUALLY_CONFIRMED ('Accepted by Owner'), never counted as Paid. Confirmation and release remain separate.

Replit hosts the app/API/worker, database and storage. Permissions, tenant filtering, version checks, hashes and allocation controls protect operations. A team-built mock payment page (/pay), bank feed and proposed Orbyte adapter are simulated; no real bank, gateway, payment or official Orbyte connection is involved.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
React/React DOM 19.2.0; Vite 7.3.7; Three.js 0.186.1	UI, frontend build and WebGL product story	MIT
Express 5.1.0; Multer 2.4.0; Zod 4.1.12; pg 8.16.3	API routing, uploads, validation and PostgreSQL access	MIT
TypeScript 5.9.3; Sharp 0.35.5; Tesseract.js 7.0.0	Typed code, image processing and CPU OCR	Apache-2.0
Playwright 1.64.1; tsx 4.20.6; concurrently 9.2.5	Browser checks, TypeScript scripts and local process orchestration	Apache-2.0 (Playwright); MIT (tsx/concurrently)
Node.js 24	Server runtime; node:test automated tests	MIT; bundled third-party components retain their notices
PostgreSQL 17; embedded-postgres 17.10.0-beta.17	Database and local development helper	PostgreSQL License; MIT (helper)
@replit/object-storage 1.0.0; @tesseract.js-data/eng 1.0.0	Private image-storage SDK and English OCR data	MIT
vLLM (RunPod worker)	Qwen model inference server	Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless / OpenAI-compatible vLLM	Receipt image -> structured fields	Metered paid/credit-funded; sponsor platform use declared
Telegram Bot API	Webhook intake, image retrieval and acknowledgements	API access; no separate paid tier recorded
Google Forms / Apps Script / Drive	Order-linked form and image forwarding	Team Google account; standard account access
KrackedDevs OpenAI-compatible API: litellm.rapidscreen.io; gpt-5.4-mini	Live advisory third reader on GPU/CPU disagreements + offline 36-image diagnostic; no field/rule/payment changes	Sponsor-provided key, capped at US$20
Replit App Storage SDK/API	Private receipt object persistence	Replit platform; US$20 credit allocation
GitHub API / GitHub	Source collaboration and repository review	Public repository; standard connected-account access
Bank, payment gateway, official Orbyte	No real external banking/payment/Orbyte API. Team-built /pay mock, synthetic bank feed and local Orbyte adapter are simulated.	Not applicable

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit deployment / Reserved VM	React build, Express API, persistent worker	Reserved VM: 0.5 vCPU / 2 GiB, about US$15/month; US$20 credit allocation. Replit-managed, region not user-selectable.

Service & Provider	Purpose in Project	Plan / Region
Replit PostgreSQL	Application database	Replit-managed database; region managed by provider
Replit private App Storage	Tenant-prefixed receipt images	Private App Storage; Replit-managed region
RunPod Serverless	vLLM/Qwen GPU inference	Serverless, 24 GB GPU; Active workers=0; idle timeout 5s. Provider-managed region.
Google Forms/Apps Script/Drive	Form hosting, submit trigger and file transfer	Team Google account; provider-managed region
GitHub	Source history, documentation and evidence	Public repository; provider-managed service
Local PostgreSQL / CPU OCR / browser WebGL	Development, tests, fallback and product story	Local compute; no cloud authentication service declared

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen2.5-VL-7B-Instruct	Qwen / RunPod vLLM	Pre-trained; primary receipt extraction; no fine-tuning	Apache-2.0
Tesseract English data 1.0.0	@tesseract.js-data/eng; Tesseract.js 7	Pre-trained OCR data; CPU reading/fallback	Package MIT; Tesseract.js Apache-2.0
gpt-5.4-mini	KrackedDevs: litellm.rapidscreen.io	Hosted pre-trained; live advisory third reader on GPU/CPU disagreement + offline diagnostic; cannot change fields/rules/payment	Provider terms; weights not distributed
Self-trained/fine-tuned models	None	No project training or fine-tuning	Not applicable

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic orders, customers, accounts, bank credits	src/db/ and src/fixtures/ in PaySah_Final	Synthetic	Team-created; repository Apache-2.0	N
36-image receipt diagnostic and expected fields	src/fixtures/; src/evaluation/compare-vlm/	Synthetic; correlated layouts; not held out	Team-created	N
81 new-value receipt images	docs/evidence/cpu-ocr/new-values/	Synthetic; same three layouts	Team-created	N
8 structured workflow fixtures / Laundry and Bakery	src/fixtures/finalist/; docs/evidence/e2e/	Synthetic; API/rule testing	Team-created	N
English OCR language data 1.0.0	@tesseract.js-data/eng	Public pre-trained language data	Package MIT	N
Real customer/bank training dataset	None; no training or fine-tuning	Not applicable	Not applicable	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
OpenAI Codex (GPT-based)	Code, tests, integration, review, UI changes and documentation	Major
Anthropic Claude (Sonnet-class models)	Code review, documentation, UI patches and final-document review	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team reviewed generated code and documentation against requirements, examined changes, and used TypeScript/build checks, isolated PostgreSQL integration tests and browser checks with synthetic records. Published 1 November validation records 179 passing tests, a successful build and 5/5 simulated intake checks. AI output is not treated as payment evidence or measured merchant benefit. Model errors, Owner override behaviour and evidence limitations are disclosed. This revision incorporates the team's final-document review; responsibility for the submitted content and signed declarations remains with the team.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Sponsor cloud platform; US$20 credit allocation	Hosts UI/API/worker, PostgreSQL and private receipt images; Reserved VM 0.5 vCPU / 2 GiB, about US$15/month
RunPod	Sponsor GPU/cloud platform	Serverless Qwen inference on 24 GB GPU; Active workers=0, idle timeout 5s; historical proof of use retained
KrackedDevs	Sponsor-provided GenAI key, capped at US$20	gpt-5.4-mini live advisory third reader through litellm.rapidscreen.io and offline 36-image diagnostic

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit and RunPod can be replaced by another Node/React/PostgreSQL host and compatible model service. CPU OCR and fixture mode work without GPU credentials, with reduced extraction capability. KrackedDevs is an optional live advisory reader for GPU/CPU disagreements; disabling it leaves the core workflow operational. It also supported offline evaluation. Replit has a US$20 credit allocation; the KrackedDevs sponsor key is capped at US$20. These allocations are not claims of

current remaining balance. RunPod usage evidence documents platform use rather than a newly redeemed credit amount.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The demo uses fictional shops, orders, receipts and bank credits. It handles login/session data, Telegram metadata, HMAC-hashed sender identifiers, receipt images and logs. Google Forms/Drive receive files; Replit stores private images and application data. RunPod receives images for extraction. The KrackedDevs endpoint (litellm.rapidscreen.io) receives disputed synthetic receipt images when enabled, only on GPU/CPU disagreement. Only synthetic samples should be submitted.

Controls include tenant-scoped access, server permissions, private objects, authenticated image delivery, SHA-256 integrity checks, no-store responses and server-side secrets. Orphan cleanup removes unreferenced objects older than 24 hours, not retained receipts. Reset preserves a 24-hour quota/idempotency ledger; older entries are removed on later reset. No automatic general retention scheduler or complete data-subject deletion workflow exists. Production privacy notices, processor terms, retention periods and deletion procedures require completion before real customer use.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Vision models can invent unreadable fields; CPU OCR misreads accounts/references and struggles with rotation or low resolution. On the same 36 synthetic images: historical Qwen2.5-VL GPU 248/252 fields; latest KrackedDevs gpt-5.4-mini 249/252; CPU OCR 224/252. Runs used different conditions and prompt/parser versions; this is not a model ranking or real-bank accuracy estimate. The gpt-5.4-mini rule run produced six false-eligible cases; no approval/release was executed.

By default PaySah warns and blocks; the Owner may override at their own responsibility ('Approve anyway', with a reason and confirmation prompt). Every override is logged as MANUALLY_CONFIRMED ('Accepted by Owner'), never counted as Paid. Independent synthetic bank evidence, duplicate controls and separate release mitigate errors. GPU timeouts use recorded CPU fallback. KrackedDevs advice cannot change fields, rules or payment. Real bank ingestion, merchant time savings, scale and current cost per receipt remain unvalidated.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[Signature]

1/11/2026

[Date]